Bernoulli-EKF on SE(3):
A Compact Multi-Object Tracker for Mobile Manipulation

Part I — One-page Algorithmic Summary

The system maintains per-object Bernoulli–Gaussian beliefs over rigid-body poses on SE(3), conditioned on RGB-D detections from a moving camera mounted on a manipulator. Each frame k ingests the SLAM base pose $T_{wb,k}$, the camera-to-base extrinsic $T_{bc,k}$, the gripper-to-base $T_{bg,k}$, the gripper finger width, and a list of detections $\{(\mathrm{pid}_d, \mathrm{label}_d, \mathrm{mask}_d, \mathrm{score}_d)\}_d$ projected through depth.

1. State. Track o stores (μ_{bo}, Σ_{bo}, r_o, ℓ_o, τ_o): $\mu_{bo} \in \mathrm{SE}(3)$ is the object pose in the *base* frame, $\Sigma_{bo} \in \mathbb{R}^{6\times6}$ is the body-frame tangent covariance at μ_{bo}, $r_o \in [0,1]$ is the existence probability, ℓ_o a label histogram, and τ_o the SAM2 perception id. *Base-frame storage is the canonical choice*: SLAM uncertainty Σ_{wb} never enters the recursion; the world view is the deterministic composition $T_{wo} = T_{wb}\,\mu_{bo}$.

2. Predict. Two motion regimes apply per track. *World-static*: $\mu_{bo,k} = u_k\,\mu_{bo,k-1}$ with $u_k = T_{wb,k}^{-1}T_{wb,k-1}$, propagated through the adjoint with phase-scheduled Q_{static}. For tracks unobserved at $k{-}1$ the covariance update is *skipped*, freezing Σ_{bo} at the last-observed keyframe; combined with the deterministic mean update this preserves $T_{wo,k} = T_{wb,k_f}\,\mu_{bo,k_f}$ for static objects under any base motion. *Rigid-attached*: held tracks evolve as $\mu_{bo,k} = \delta_{bg,k}\,\mu_{bo,k-1}$ with phase-aware Q_{manip}. At the holding→idle (release) transition, a one-shot gravity-aware predict overwrites the just-released track's world-frame pose and covariance using voxel-occupancy-driven drop height and a per-label restitution / footprint table.

3. Detection conditioning. Subpart suppression voxelises masks through depth at 2 cm and absorbs detections whose voxel sets are mostly contained in another's; centroid back-projection drops masks without valid depth.

4. Association. Translation-only Mahalanobis with a χ^2_3 gate at $G_{\mathrm{out},t}$=21.108 and R_{cam}=$(0.02\,\mathrm{m})^2 I$, plus a hard Euclidean residual cap (default 0.30 m) that catches inflated covariances. Hungarian on the gated cost; a SAM2-id continuity bonus α=4.4 (in χ^2 units) breaks ties. A configurable label gate runs in either hard mode (mismatched labels become infeasible) or soft mode (label mismatch penalty β + score weight γ).

5. Update. For matched pairs on the held track, two safety prefilters guard the Joseph step: a held-prefilter (≤ 0.25 m from the end-effector) and an innovation clamp (≤ 0.20 m from μ_w). The Joseph step runs on the *translation block only*; the rotation block and cross-covariance are zeroed (a translation-only observation carries no rotation information). The world-frame view used by all downstream consumers is the canonical composition $T_{wo} = T_{wb}\,\mu_{bo}$ with $\Sigma_{wo} = \mathrm{Ad}(\mu_{bo}^{-1})\,\Sigma_{wb}\,\mathrm{Ad}(\mu_{bo}^{-1})^\top + \Sigma_{bo}$.

6. Birth and prune. A confirm-3 / score-floor / border-margin / near-live admission chain controls births. Survivors mint a fresh oid, run ICP once at admission, and seed $r_0 = \lambda_b \hat{s}_d/(\lambda_b \hat{s}_d + \lambda_c)$. Tracks prune when $r_o < 10^{-3}$, except the held seed is floored at r=0.5.

7. Self-merge. Same-label tracks within 5 cm fuse via information sum, except pairs in the relation-protected set (the scene graph asserts they are physically distinct). The held seed is always the keeper.

8. Relations and held-set expansion. A VLM/REST backend computes in/on edges at first call, release transitions, new-oid arrivals, and every 90 frames. The parent matrix is converted to edges with the convention **parent = supported, child = supporter**; a per-edge EMA (α=0.3, threshold 0.5) smooths flicker. The held set is the closure of {held seed} under in/on edges; after each self-merge, EMA keys are remapped (drop $\rightarrow$ keep) so the closure stays valid.

Part II — Detailed Algorithmic Exposition

This part is purely algorithmic; it carries no source-file references. The algorithm-to-code map is in Part III.

II.1 State

$$\mathcal{S}_o = (\mu_{bo} \in \mathrm{SE}(3),\ \Sigma_{bo} \in \mathbb{R}^{6\times 6}_{\succeq 0},\ r_o \in [0,1],\ \ell_o,\ \tau_o \in \mathbb{Z},\ \text{age},\ \text{frames since obs}).$$

Body-frame tangent at μ_{bo}: a perturbation $\xi \in \mathfrak{se}(3)$ acts as $\mu' = \mu\ \mathrm{Exp}(\xi)$ (right-multiply). Ordering $\xi = (\xi_v; \xi_\omega)$ — *translation first, then rotation.* The adjoint

$$\mathrm{Ad}(T) = \begin{bmatrix} R & [t]_\times R \\ 0 & R \end{bmatrix}$$

moves a tangent vector through $T = (R, t)$.

II.2 Predict

World-static. A world-static object satisfies $T_{wb,k}\,\mu_{bo,k} = T_{wb,k-1}\,\mu_{bo,k-1}$, hence $\mu_{bo,k} = u_k\,\mu_{bo,k-1}$ with $u_k = T_{wb,k}^{-1} T_{wb,k-1}$.

The covariance recursion branches on whether the track was observed at $k-1$:

$$\Sigma_{bo,k} \;=\; \begin{cases} \Phi\big(\mathrm{Ad}(u_k)\,\Sigma_{bo,k-1}\,\mathrm{Ad}(u_k)^\top + Q_{\text{static}}\big) & \text{observed at } k-1, \\ \Sigma_{bo,k-1} & \text{unobserved at } k-1 \text{ (frozen keyframe).} \end{cases} \tag{1}$$

The mean update $\mu_{bo,k} = u_k \mu_{bo,k-1}$ runs in *both* branches, deterministically. Q_{static} is scheduled on frames since the last observation: $1\mathrm{e}{-6}\, I_6$ (idle, < 5), $1\mathrm{e}{-5}\, I_6$ (< 50), $1\mathrm{e}{-8}\, I_6$ thereafter. Φ is the optional covariance saturation

$$\Phi(P) \;=\; \min\!\big(1,\ \mathrm{tr}\, P_{\max}/\,\mathrm{tr}\, P\big)\, P, \tag{2}$$

inactive when no cap is configured (the default). The Bernoulli predict on existence is

$$r_{k|k-1} = p_s\, r_{k-1|k-1}, \qquad p_s = 1 \text{ default.} \tag{3}$$

Why the unobserved branch freezes Σ_{bo} (the keyframe mechanism). The Ad-conjugate transport carries a cross-coupling $[t(u_k)]_\times P_{rr}[t(u_k)]_\times^\top$ that pumps rotation uncertainty into the translation block each frame. For tracks with high σ_{yaw} (e.g. a freshly-released object with $\sigma_{yaw} = \pi$ from the gravity prior, (17)), the per-frame contribution $\Delta x^2{\cdot}\pi^2$ compounds catastrophically (order of one metre over a few dozen frames). The keyframe mechanism exploits the algebraic identity that, telescoping the deterministic mean update over K frames since the last observation at k_f,

$$\mu_{bo,K} \;=\; u_K \cdots u_{k_f+1}\, \mu_{bo,k_f} \;=\; T_{wb,K}^{-1}\, T_{wb,k_f}\, \mu_{bo,k_f}. \tag{4}$$

Composing with the current SLAM pose then gives $T_{wo,K} = T_{wb,K}\mu_{bo,K} = T_{wb,k_f}\mu_{bo,k_f}$ — *constant* for a static object, exactly as required. The keyframe is not stored explicitly; $\Sigma_{bo,K} = \Sigma_{bo,k_f}$ is preserved by skipping the covariance update. World composition (§II.5, (9)) then uses the *current* $\Sigma_{wb,K}$ for the lever-arm contribution, so the world-frame uncertainty grows only via the SLAM marginal — never via per-frame predict drift.

Rigid-attached. For tracks in the held set $\mathcal{H}_k$, $\mu_{bo,k} = \delta_{bg,k}\, \mu_{bo,k-1}$ with $\delta_{bg,k} = T_{bg,k}T_{bg,k-1}^{-1}$. A phase-aware Q_{manip} applies:

- Seed during grasping/releasing: diag($(0.02)^2{\times}3$, $(0.10)^2{\times}3$) — physical grip slack.
- Seed during holding: diag(2.5e$-$4×3, 1e$-$3×3) — settled grip.
- Transitive members of the held set (e.g. items resting on a held tray): always the holding row — they don't deform when the gripper closes around the seed.

The held branch runs the full Ad-conjugate (no keyframe-mechanism shortcut), so the same translation/rotation cross-pumping applies — bounded in practice by $\|t(\delta_{bg,k})\| \ll 1$ but *not* eliminated. Edge cases with $\sigma_{yaw} \to \pi$ inside the holding window (re-grasps, release transitions) remain a known caveat.

II.3 Detection conditioning

Subpart suppression. Voxelise each mask through depth at 2 cm; for ordered pair (a, b) with $|V_b| \le |V_a|$, if $|V_b \cap V_a|/|V_b| > 0.8$, absorb b into a (label histograms merge). This catches SAM mask splits and re-prompts.

Centroid validity. $c_d^{\text{cam}} = \frac{1}{|m_d \cap \text{valid-depth}|} \sum_{p \in m_d} \text{back\_project}(p, \text{depth}, K)$. If the mask has no valid-depth pixels, d is dropped.

II.4 Association

For each (o, d):

$$\begin{aligned}
t_{bo}^{\text{meas}} &= T_{bc}\big[c_d^{\text{cam}}; 1\big]_{1:3} \\
R_{bo}^t &= R_{bc}\, R_{cam}\, R_{bc}^\top, \quad R_{cam} = \text{diag}((0.02\,\text{m})^2{\times}3) \\
\nu_{od} &= t_{bo}^{\text{meas}} - \mu_{bo}[1{:}3, 4], \quad S_{od} = P_{tt,o} + R_{bo}^t \\
d_{t,od}^2 &= \nu_{od}^\top S_{od}^{-1} \nu_{od}.
\end{aligned}$$

Cost. Two modes, selected by a label-enforcement flag:

$$C_{od} \;=\; d_{t,od}^2 \;-\; \alpha_{\text{SAM2}}\, \mathbf{1}[\tau_o = \text{pid}_d,\ \tau_o \ge 0] \;+\; \beta\, \mathbf{1}[\hat{\ell}_d \notin \mathcal{H}_o] \;+\; \gamma\, \max(0,\, 1 - \hat{s}_d). \tag{5}$$

Hard label gate (paper baseline): $\beta = \gamma = 0$, and any cell with mismatched label is forced to a sentinel infeasible value. **Soft mode**: β is a configurable label penalty (default 0.0; the production driver sets 6.0) and γ is a configurable score weight (default 0.0; production sets 2.0). $\alpha_{\text{SAM2}} = 4.4$ in either mode.

Label-purity gate (soft mode only). The membership test $\hat{\ell}_d \in \mathcal{H}_o$ is not pure dictionary membership; a label is "meaningfully in the history" iff it has ≥ 5 observations *and* $\geq 5\%$ of the total. This rejects stale trace-amount labels (e.g. 3 obs against 53 990) that would otherwise bypass the β penalty.

Outer gates. The Hungarian solver minimises $\sum C_{o\sigma(o)}$. Two gates apply:

1. Probabilistic χ^2_3: $d^2_{t,od} > G_{\text{out},t} = 21.108$ (i.e. at 99.97%) $\Rightarrow$ cell infeasible.
2. Hard Euclidean residual cap: $\|\nu_{od}\|_2$ above a configurable threshold (default 0.30 m; `None` disables) $\Rightarrow$ cell infeasible. This catches the inflated-covariance pathology where, after a long miss run with $\sigma_t \approx 0.4$ m, an 0.86 m residual gives $d^2_t \approx 17 < 21.108$ and the χ^2 gate becomes meaningless.

$$\sigma^* = \arg\min_{\sigma} \sum_{o} C_{o\,\sigma(o)}, \qquad \text{drop } (o,d) \text{ if } C_{od} \text{ is infeasible.} \tag{6}$$

II.5 Update

Held-track prefilters. For each accepted match (o,d) where o is the held seed:

- Held-prefilter: $\|c^w_d - (T_{wb}T_{bg})_{1:3,4}\|_2 \leq 0.25$ m.
- Innovation clamp: $\|c^w_d - (T_{wb}\mu_{bo})_{1:3,4}\|_2 \leq 0.20$ m.

A failure marks the detection consumed (so it does not re-birth) and skips the EKF step; the rigid-attach predict carries the track.

Joseph step (translation-only). Let $K_{tt} = P_{tt}\, S^{-1}$ (3×3):

$$\begin{aligned}
\delta_t^{\text{base}} &= K_{tt}\,\nu \\
\mu_{bo}[1{:}3,4] &\leftarrow \mu_{bo}[1{:}3,4] + \delta_t^{\text{base}} \\
P_{tt}^{\text{post}} &= (I - K_{tt})\,P_{tt}\,(I-K_{tt})^\top + K_{tt}\,R^t_{bo}\,K_{tt}^\top \\
\Sigma_{bo}^{\text{post}} &= \text{block}-\text{diag}\big(P_{tt}^{\text{post}},\ P_{rr}\big).
\end{aligned}$$

The cross-covariance is zeroed: a translation-only observation provides no information about rotation (we set $R_{rot} = \infty$), so the honest posterior is block-diagonal. An earlier version preserved the prior cross-covariance, which broke the Schur condition $P_{rr} - P_{rt}^\top P_{tt}^{-1} P_{rt} \succeq 0$ over many updates; the current version is PSD by construction, with an eigenvalue clip as a belt-and-suspenders.

For pairs where ICP succeeded, an alternative full-SE(3) update via the Joseph form on the 6-D innovation lifts the camera-frame measurement to base frame:

$$T_{bo}^{\text{meas}} = T_{bc}\,\hat{T}^{co}, \qquad R^{bo} = \text{Ad}(T_{bc})\,\hat{R}^{co}\,\text{Ad}(T_{bc})^\top, \tag{7}$$

followed by IEKF inner iterations and Huber inner / outer reweighting ((10)). The EKF likelihood used by (13) is

$$\log\mathcal{L} = -\tfrac{1}{2}\big(\nu^\top S^{-1}\nu + \log\det(2\pi S)\big), \tag{8}$$

computed in log space to avoid underflow at $\log\mathcal{L} \sim -50$.

World-frame composition. The canonical world-frame view consumed by every downstream client (relations, slow tier, gravity-predict input, visualizer) is

$$\mu_{wo} = T_{wb}\,\mu_{bo}, \qquad \Sigma_{wo} = \text{Ad}(\mu_{bo}^{-1})\,\Sigma_{wb}\,\text{Ad}(\mu_{bo}^{-1})^\top + \Sigma_{bo}. \tag{9}$$

Σ_{wb} is read from the current SLAM marginal at query time, so the world-cov tracks any SLAM drift / loop-closure shift even when no new object observation has arrived. Combined with the keyframe-mechanism freeze of Σ_{bo} ((1)), an unobserved static track's world cov grows *only* via the Σ_{wb} lever — the user-intended behaviour.

World-frame overwrite (release hook). A pose-overwrite primitive forces the EKF state for one oid from a world-frame (T_{wo}, P_{wo}) input. It is invoked at the release transition to install the gravity-aware landing prior ((17)); the body-frame storage becomes $\mu_{bo} = T_{wb}^{-1} T_{wo}$ and $\Sigma_{bo} = P_{wo}$ (the right-trivialised tangent transport reduces to the identity at the new μ_{bo}). After the overwrite, the keyframe-mechanism freeze in (1) keeps the gravity-supplied covariance ($\sigma_{yaw} = \pi$, $\sigma_{xy} \sim 5$–10 cm) intact across subsequent unobserved frames.

Per-track observation chain (diagnostic). Each track also accumulates an append-only chain $\mathcal{H}^{(i)} = [(\,k, \hat{T}_k^{co}, \hat{R}_k^{co}\,), \ldots]$ of camera-frame ICP measurements. **The chain is NOT the canonical world-frame source** — that role belongs to the world composition (9). The chain is computed in parallel by the visualizer for cross-checking and is the natural substrate for a future loop-closure-aware smoother.

Huber reweight.

$$w(d^2; G_{\text{in}}, G_{\text{out}}) = \begin{cases} 1 & d^2 \le G_{\text{in}}, \\ G_{\text{in}}/d^2 & G_{\text{in}} < d^2 \le G_{\text{out}}, \\ 0 & d^2 > G_{\text{out}}. \end{cases} \tag{10}$$

Defaults: $G_{\text{in},t} = 7.815$ (χ^2_3 at 95%), $G_{\text{out},t} = 21.108$.

II.6 Birth and prune

Birth gates:

1. Border: bbox not within 2 px of the image edge.
2. Confirm-k: tracker-side pending counter $\ge k = 3$ (counts the tracker's unmatched-streak frames, not perception's n_{obs} which resets on SAM2 id reissue).
3. Score floor: rolling-max score ≥ 0.20.
4. Near-live: world-frame centroid > 0.05 m from any same-label live track. For the held oid the radius widens to 0.25 m and T_{we} is checked as a secondary anchor.
5. TTL: pending entries that go 30 frames without a matching arrival are evicted (one second at 30 Hz).

Survivors mint a fresh oid, run ICP once on admission, and seed

$$r_0 \;=\; \frac{\lambda_b\, \hat{s}_d}{\lambda_b\, \hat{s}_d + \lambda_c}. \tag{11}$$

Birth covariance defaults to

$$P_{\text{new}} = \hat{R}^{co} \quad \text{(from ICP, by default)} \tag{12}$$

and falls back to $\operatorname{diag}((5\,\text{cm})^2{\times}3, (0.05\,\text{rad})^2{\times}3)$ when the ICP-derived form is disabled.

Existence updates. For matched pairs

$$r' = \frac{(1 - p_s) + p_s\, p_d\, \mathcal{L}\, r}{1 - p_s + p_s\, r}, \tag{13}$$

and for missed tracks (with $\tilde{p}_d = p_d\, p_v$)

$$r' = \frac{(1 - \tilde{p}_d)\, r}{1 - \tilde{p}_d\, r}, \tag{14}$$

with p_v the visibility probability of (16).

Prune. $r_o < 10^{-3}$ deletes the track, EXCEPT the held seed, whose existence is floored at 0.5 (survives long held-occlusion).

II.7 Self-merge

Inputs: the held seed s, and protected pairs $\mathcal{P}$ derived from the relation edge set $\mathcal{E}$ (§II.9). The parent/child convention is *parent = supported, child = supporter*, so a relation (p, c, on) contributes the unordered pair $(\min(p, c), \max(p, c))$ to $\mathcal{P}$:

$$\mathcal{P} = \big\{\, (\min(p, c), \max(p, c)) \;:\; (p, c, \cdot) \in \mathcal{E} \,\big\}.$$

Same-label pairs in $\mathcal{P}$ are protected from collapse — the scene graph asserts they are physically distinct (item-on-tray, etc.). Information-sum fusion (with the keep mean as linearisation point) is

$$\begin{aligned} P_{\text{new}}^{-1} &= P_{\text{keep}}^{-1} + P_{\text{drop}}^{-1}, \\ \mu_{\text{new}} &= \mu_{\text{keep}}\, \operatorname{Exp}\!\big(P_{\text{new}}\, P_{\text{drop}}^{-1}\, \operatorname{Log}(\mu_{\text{keep}}^{-1} \mu_{\text{drop}})\big). \end{aligned} \tag{15}$$

Algorithm 1 Self-merge pass.

for each pair (o_i, o_j) with $\ell_i = \ell_j$ **do**
  if $(\min(o_i, o_j), \max(o_i, o_j)) \in \mathcal{P}$ **then continue**
  end if
  $d \leftarrow \|\mu_w(o_i) - \mu_w(o_j)\|_2$
  if $d > 0.05$ **then continue**
  end if
  Push (d, d_t^2, o_i, o_j) as candidate.
end for
Sort candidates by d ascending.
for each (d, d_t^2, o_i, o_j) **do**
  if o_i or o_j already absorbed **then continue**
  end if
  Choose keep$\leftarrow s$ if $s \in \{o_i, o_j\}$, else higher-r, else lower oid.
  Information-sum fuse μ, Σ; merge label histograms / SAM2 id / T_{oe} / r.
  Append {keep, drop, d, d_t^2} to merges.
end for

The protected-pairs gate is what stops a same-label item-on-tray pair from collapsing once their world centroids drift within 5 cm under occlusion bias.

II.8 Visibility

For each track i, sample object-surface points $\{x_j^{(i)}\}_{j=1}^{M}$ on a Fibonacci sphere of radius r_{obj} around μ_{bo} (the paper specifies projecting the per-object TSDF shape summary; the current implementation uses the spherical fallback). Lift each through T_{bc} into camera, depth-test against the current depth image:

$$p_v^{(i)} \;=\; \frac{\#\{j : x_j^{(i)} \text{ in-frustum and passes depth test}\}}{\#\{j : x_j^{(i)} \text{ in-frustum}\}} \;\in\; [0,1]. \tag{16}$$

$p_v = 1$ when no in-frustum sample is occluded; $p_v = 0$ when fully out-of-frustum or fully occluded. The continuous probability composes naturally with (14); the four-state machine (present / occluded / absent / not-in-view) collapses into a single weight on the miss-frame likelihood.

II.9 Relations

Trigger gate. The relation backend fires at first call, at release transitions, on new-oid arrivals, or every 90 frames. The grasp-transition trigger was removed (the backend at grasp onset has incomplete information).

Backend. On a triggered frame, the orchestrator builds normalised bboxes per oid (from the detection-to-oid map) and calls a VLM (LLM-based) or REST relation client. The backend returns $p \in [0,1]^{N\times N}$ with $p_{ij} =$ "j rests on i". A disk cache keyed by $(\text{frame}, N, \text{rounded bboxes})$ is read on retry.

Convention swap. Held-set expansion uses parent = supported, child = supporter. The backend's (i, j) encodes the opposite (j rests on i). Edge emission swaps:

$$\text{edge} = \text{RelationEdge}(\text{parent} = j, \text{child} = i, \text{type} = \text{on}, \text{score} = p_{ij}).$$

EMA. For each key $k = (\text{parent}, \text{child}, \text{type})$:

$$\text{ema}_k(t) = \alpha\, \text{raw}_k(t) + (1-\alpha)\, \text{ema}_k(t-1), \quad \alpha = 0.3.$$

Edges emit when ema ≥ 0.5 and prune when ema ≤ 0.01. Half-life is $\sim$2 trigger calls $\sim$180 frames at 30 Hz.

Remap after merges. When self-merge produces $\{(\text{drop}_i, \text{keep}_i)\}$, every EMA key and emitted edge has parent/child rewritten via the closure of `drop_to_keep`. Self-loops drop, duplicate keys merge by max.

II.10 Held-set expansion

$S \leftarrow \{s\}$
for iter $= 1 \ldots 8$ **do**
    changed $\leftarrow$ false
    for each edge $(p, c, t) \in \mathcal{E},\ t \in \{\text{in}, \text{on}\}$ **do**
        if $c \in S$ and $p \notin S$ **then** $S \leftarrow S \cup \{p\}$; changed$\leftarrow$true
        end if
    end for
    if not changed **then break**
    end if
end for
return S

The caller post-filters S against currently-live oids to drop references to pruned tracks the EMA has not yet decayed.

II.11 Gravity-aware predict at release

When the gripper releases an object, the rigid-attachment predict no longer applies and the static predict over-states the prior (the object physically falls). This subsystem replaces the EKF mean and covariance for the just-released oid with the post-settling prediction in a single step at the RELEASING $\to$ IDLE transition.

Voxel observability. A dense 3-state grid (UNSEEN / EMPTY / OCCUPIED) plus per-voxel hit/pass counters with hysteresis (≥ 2 confirming hits and ≥ 3 pass-through rays before transition) is updated each frame from depth: each pixel's ray marks voxels traversed before the depth hit as pass-pending and the voxel at the hit as hit-pending. Default workspace $5\,\text{m} \times 5\,\text{m} \times 3\,\text{m}$ at $5\,\text{cm}$ voxel size.

Object dynamics property. A per-label table maps each detection label to (e, μ, shape, r_{obj}). Restitution defaults from agricultural / engineering literature; a generic fallback with $e = 0.40$, $r = 0.05$ applies to unknown labels. A perception-side override hook lets a learned estimator supersede the table per-detection.

Parametric dispersion model. For drop height h (resolved via voxel-driven raycast, below):

$$\begin{aligned}
v_{\text{impact}} &= \sqrt{2gh}, \qquad t_{\text{fall}} = \sqrt{2h/g}, \\
\sigma_{\text{bounce}} &= \varepsilon_{\text{rough}}\, v_{\text{impact}} \big/ \max(0.1,\, 1 - e), \\
\sigma_{\text{shape}} &= r_{\text{obj}}\, \text{footprint_factor}(\text{shape}), \\
\sigma_{\text{rel-v}} &= \tfrac{1}{2}\, \|\boldsymbol{v}_{\text{rel,xy}}\|\, t_{\text{fall}}, \\
\sigma_{xy}^2 &= \sigma_{\text{bounce}}^2 + \sigma_{\text{shape}}^2 + \sigma_{\text{rel-v}}^2, \\
\sigma_z &= \tfrac{1}{2}\, r_{\text{obj}}\, (1 - e/2), \\
\sigma_{\text{yaw}} &= \pi\, \big(1 - \exp\!\big(-h(1 - e^2)/r_{\text{obj}}\big)\big).
\end{aligned} \tag{17}$$

Default $\varepsilon_{\text{rough}} = 5{\cdot}10^{-3}\,\text{m} / (\text{m s}^{-1})$; footprint factor by shape: spherical 0.25, cylindrical 0.50, box 0.70, irregular 1.00. Higher restitution increases σ_{bounce} — physically, a rubber ball bounces several times each accruing scatter, while a clay ball lands once and stays.

Voxel-driven h branch. The downward raycast classifies the column into one of four states; each adjusts h, landing z, and the σ_z floor:

- `hit_occupied`: clean surface at $h^\star$; $h = h^\star$, σ_z from (17).
- `mixed_unseen`: surface known but UNSEEN voxels above it; $h = h^\star$, σ_z extra-inflated by $(h_u - h^\star)/2$ to reflect path uncertainty.
- `all_unseen`: surface unknown; $h = (h_u + h_{\text{floor}})/2$ (*midpoint*, not worst case), $\sigma_z = (h_u - h_{\text{floor}})/2$.
- `all_empty`: column observed empty all the way to the workspace floor; $h = h_{\text{floor}}$, $\sigma_z = r_{\text{obj}}$.

The midpoint choice for `all_unseen` gives the EKF a useful prior that subsequent observations can correct quickly, instead of pushing the mean to an extreme that is unlikely to be right.

Trigger. The release predict fires once at $\{\text{Holding}, \text{Releasing}\} \to \text{Idle}$. A toggle flag (default on) gates the hook; with the voxel grid unset, the hook short-circuits and the legacy static-Q-only release behaviour is preserved exactly. Subsequent frames see the released track as unobserved, so (1)'s keyframe freeze preserves the gravity prior (in particular $\sigma_{yaw} = \pi$ does not pump σ_{xy} via cross-coupling).

II.12 Gripper phase FSM

Four phases: Idle → Grasping → Holding → Releasing → Idle. **Project assumption:** the gripper begins fully open; ANY closing of the fingers registers as a grasp action. The state machine therefore detects "closed" as a drop ≥ 0.005 m below the rolling open baseline (5 mm captures any non-trivial finger motion while ignoring proprioception jitter), in addition to the absolute thresholds. Drivers:

- Width threshold crossings with hysteresis (closed 0.025 m, open 0.040 m, drop-below-baseline 0.005 m).
- Stability test: last 5 widths span < 0.01 m.
- Min-frames timer (5 frames) holds the transition phases through brief instability.

At the Idle → Grasping edge, the held oid is selected by ranking detections by inside-gripper-volume containment fraction (=count_inside/total_pts); ties break by smallest mask area. A fallback returns the nearest live track within 0.30 m of T_{we}. Self-merges propagate via an apply-merges hook: if the dropped id was the held seed, it switches to the keep id.

II.13 Frame-by-frame algorithm

Algorithm 2 One frame of the Bernoulli-EKF tracker.

Inputs: T_{wb}, T_{bc}, T_{bg}, w, RGB, depth, detections.
$g \leftarrow$ GripperPhaseTracker.step$(w, \ldots)$
Maybe call relations backend (§II.9 trigger).
$\mathcal{H} \leftarrow$ Expand$(g.\text{held}, \mathcal{E}_t) \cap$ live oids
$\mathcal{U} \leftarrow \{o : \text{frames since obs}[o] > 0\}$ ▷ keyframe-frozen set
predict_static(skip $= \mathcal{H}$, unobserved $= \mathcal{U}$); rigid_attachment_predict(each $o \in \mathcal{H}$)
suppress_subpart_detections; centroid_cam_from_mask; gather detections
hungarian_associate $\rightarrow$ matches, unmatched_tracks, unmatched_dets
for each match (o, d) **do**
    if $o = g$.held and (held-prefilter or innov-clamp triggers) **then** skip update
    end if
    update_observation_centroid; update r_o via match log-likelihood
end for
for each unmatched track o **do**
    Update r_o via miss equation; floor at $r_{\text{held floor}}$ if $o = g$.held
end for
for each unmatched detection d **do**
    if birth-admissible(d) and not near-live(d) **then**
        Mint new oid, run ICP once, set r_0 per (11)
    end if
end for
Prune oids with $r_o < r_{\min}$ (skip held seed)
$\mathcal{P} \leftarrow \{(\min(p, c), \max(p, c)) : (p, c, \cdot) \in \mathcal{E}_t\}$
Self-merge pass with held$= g$.held and protected$= \mathcal{P}$
g.apply-merges(merges); relation pipeline.remap-after-merges(merges)
Maybe gravity-predict (release transition: overwrite the just-released oid via (17))

II.14 Numerical defaults summary

R_{cam}	$\mathrm{diag}((0.02\,\mathrm{m})^2)$	centroid measurement
Q_{static}	1e−6 → 1e−5 → 1e−8	schedule on frames since obs
$Q_{\text{manip seed}}$ (grasping/releasing)	$\mathrm{diag}((0.02)^2{\times}3, (0.10)^2{\times}3)$	per frame
$Q_{\text{manip seed}}$ (holding)	$\mathrm{diag}(2.5\mathrm{e}{-}4{\times}3, 1\mathrm{e}{-}3{\times}3)$	per frame
$Q_{\text{manip transit}}$	holding row above	transitive held members
$G_{\text{out},t}$	21.108	χ^2_3 at 99.97%
$G_{\text{in},t}$	7.815	χ^2_3 at 95% Huber inner
α_{SAM2}	4.4	continuity bonus
max_residual	0.30 m	hard Euclidean residual cap (None disables)
P_{max}	none	covariance saturation cap (no cap default)
birth-min-dist	0.05 m	default near-live radius
birth-border-margin	2 px	image-edge margin
birth-pending-TTL	30 frames	pending-birth TTL (∼1 s @ 30 Hz)
held-birth-radius	0.25 m	wider for the held oid
held-meas-radius	0.25 m	end-effector prefilter
held-meas-innov-max	0.20 m	innovation clamp
r_{min}	10^{-3}	prune threshold
$r_{\text{held floor}}$	0.50	held existence floor
birth-confirm-k	3	tracker-side unmatched-streak count
birth-score-min	0.20	rolling-max score floor
self-merge-trans	0.05 m	Euclidean merge gate
dedup-voxel-size	0.02 m	subpart suppression voxel size
dedup-containment-thresh	0.8	subpart suppression threshold
β (label penalty)	0.0 default	soft mode; production sets 6.0
γ (score weight)	0.0 default	soft mode; production sets 2.0
closed/open width	0.025 m, 0.040 m	FSM hysteresis
close-delta	0.005 m	drop-below-baseline = grasp
grasp-radius	0.30 m	nearest-track gate at grasp onset
EMA α, threshold	0.3, 0.5	relation smoothing
relation-every-n-frames	90	periodic backend tick
gravity-predict	on	one-shot release-time gravity hook
workspace-floor-z	−1.0 m	column terminator for raycast-down
$\varepsilon_{\text{rough}}$	$5{\cdot}10^{-3}$	bounce-roughness coefficient
voxel grid (default)	5 cm at 5×5×3 m	observability grid
$n^{\text{hit}}_{\text{min}}$, $n^{\text{pass}}_{\text{min}}$	2, 3	voxel hysteresis thresholds

---

Part III — Algorithm-to-Code Map

Each line below cites exactly one source-file path. Where one file hosts multiple algorithmic items, the path repeats across multiple lines, each line scoping a distinct role.

Public API and configuration

- Public five-method surface (`detect`, `step`, `get_scene`, `get_points`, `smooth`): `ekf_tracker/api.py`.

- Bernoulli/EKF/birth thresholds and gates (§II.4, II.6 numerics): `ekf_tracker/config.py`.
- Live OWL+SAM2 streaming detection pipeline behind `detect`: `ekf_tracker/perception_pipeline.py`.

State

- Per-object Bernoulli-Gaussian belief and storage: `ekf_tracker/state/gaussian_state.py`.
- Existence-probability arithmetic ((3), (13), (14), (11)): `ekf_tracker/state/bernoulli.py`.
- SE(3) tangent algebra, Joseph form, Huber reweight, Q schedule: `utils/ekf_se3.py`.
- Lift-to-base and information-sum primitives used by ICP and self-merge: `utils/object_belief.py`.
- Append-only per-track ICP-measurement chain (diagnostic, smoother substrate): `ekf_tracker/state/obs_chain.py`.
- RBPF backend state (legacy fast tier; not the canonical path): `ekf_tracker/state/rbpf_state.py`.

Predict

- World-static and rigid-attachment per-track predict ((1), (4)): `ekf_tracker/state/gaussian_state.py`.
- Phase-aware Q schedule and adjoint covariance transport: `utils/ekf_se3.py`.
- Step orchestration and held-set wiring: `ekf_tracker/gaussian_ekf_tracker.py`.
- Two-tier orchestrator gluing fast tier and slow tier: `ekf_tracker/orchestrator_gaussian.py`.

Detection conditioning

- Voxel-containment subpart suppression and label-histogram merge: `perception/det_dedup.py`.
- Mask cleaning, depth back-projection, centroid validity gate: `perception/icp_pose.py`.

Association

- Hungarian solver, χ^2_3 gate, residual cap, label penalty ((5), (6)): `perception/association.py`.
- Default values for α_{SAM2}, G_{out}, β, γ and the label-enforcement flag: `ekf_tracker/config.py`.

Update

- Translation-only Joseph step on the centroid measurement (§II.5): `ekf_tracker/state/gaussian_state.py`.
- Full-SE(3) Joseph form and IEKF inner iterations (used when ICP succeeds): `utils/object_belief.py`.
- EKF predict, EKF update on SE(3), Huber reweight ((10)): `utils/ekf_se3.py`.
- World-frame composition μ_{wo}, Σ_{wo} ((9)): `ekf_tracker/state/gaussian_state.py`.
- World-frame pose-and-cov overwrite (release-hook primitive): `ekf_tracker/state/gaussian_state.py`.

Birth and prune

- Confirm-k / score / border / TTL admission gate: `ekf_tracker/birth_gate.py`.
- Near-live-track world-frame proximity gate: `perception/birth_gating.py`.
- Birth existence probability r_0 ((11)): `ekf_tracker/state/bernoulli.py`.
- One-shot ICP at admission (with mask cleaning and depth filter): `perception/icp_pose.py`.

Self-merge

- Self-merge pass (candidate enumeration, sort, fuse) and merge-event emission: `ekf_tracker/gaussian_ekf_tracker.py`.
- Information-sum fusion primitive ((15)): `ekf_tracker/state/gaussian_state.py`.

Visibility

- Fibonacci-sphere sampling and depth-test visibility p_v ((16)): `perception/visibility.py`.

Relations

- Trigger gate, backend dispatch, edge emission, drop-to-keep remap: `ekf_tracker/relations/relation_orchestrator.py`.
- LLM and REST clients with disk-cached prompts/responses: `ekf_tracker/relations/relation_client.py`.
- Per-edge EMA smoother (§II.9): `ekf_tracker/relations/relation_filter.py`.
- Held-set closure under in/on edges (§II.10) and trigger predicate: `ekf_tracker/relations/relation_utils.py`.

Gripper FSM

- Four-phase FSM with hysteresis, stability, and min-frame timer: `utils/gripper_state.py`.
- Held-oid selection at the grasp edge (containment-fraction ranking): `ekf_tracker/manipulation/grasp_owner_detector.py`.
- Phase + held-set wiring into the tracker step: `ekf_tracker/manipulation/gripper_state_inferrer.py`.
- Gripper finger geometry primitives shared by FSM and visualizer: `utils/gripper_geometry.py`.

Gravity-aware release

- Voxel observability grid and downward raycast classification: `perception/voxel_observability.py`.
- Per-label dynamics table $(e, \mu, \text{shape}, r_{\text{obj}})$ with default fallback: `utils/object_dynamics.py`.
- Parametric dispersion model ((17)) and landing-pose synthesis: `ekf_tracker/manipulation/gravity_predict.py`.
- One-shot release-transition hook calling the overwrite primitive: `ekf_tracker/api.py`.

Driver, harness, slow tier

- Canonical instrumented fast-tier driver with debug capture: `scripts/visualize_ekf_tracking.py`.
- Fast-tier orchestration (per-frame step body, debug output): `ekf_tracker/gaussian_ekf_tracker.py`.
- Slow tier (factor-graph smoother) and its trigger schedule: `ekf_tracker/factor_graph.py`.
- SLAM marginal interface and base-pose adapters: `utils/slam_interface.py`.
- Robot kinematics and end-effector geometry (Fetch baseline): `utils/robot_models/fetch.py`.
- Deep camera-pose refiner used only when explicitly enabled: `perception/camera_pose_refiner.py`.
- Adaptive measurement-noise kernel for centroid likelihood: `perception/adaptive_kernel.py`.

Tests and diagnostics

- Sole quantitative regression (parity against cached detections): `tests/parity_apple_drop.py`.
- Live OWL+SAM2 detect → EKF step smoke test: `tests/integration/test_detect_step_pipeline.py`.
- Per-API mp4 renderer (5 separate videos, one per public method): `tests/diagnostics/render_api_videos.py`.
- API demo with overlay panels (single-trajectory): `tests/demo_apis.py`.

Part IV — Reviewer Critique

The audit underlying this part is summarised by category. Citations use `file.py:line` for evidence. The critique is intended for a top-tier robotics venue review and is intentionally blunt.

IV.A Single-trajectory hardcoding

The runnable pipeline carries trajectory-specific anchors in its production sources, not just in scripts:

- `ekf_tracker/manipulation/gravity_predict.py:46-47` sets the bounce-roughness coefficient $\varepsilon_{\text{rough}} = 5{\cdot}10^{-3}$ with the inline comment "tune empirically on the apple_drop trajectory".
- `ekf_tracker/gaussian_ekf_tracker.py:82` justifies a held-set rule with a specific frame range of `apple_drop fr 269-273`.
- `ekf_tracker/gaussian_ekf_tracker.py:924` cites a specific `apple_in_the_tray` frame as the failure case the fix is keyed to.
- `ekf_tracker/state/gaussian_state.py:612` bounds drift by citing "apple_in_the_tray oid 3, frames 290-330".
- `perception/icp_pose.py:272` defends a heuristic with the apple_in_the_tray frame-430 case.
- `utils/fetch_kinematics.py:17` calibrates camera-to-base rotation against `/tf` from the `apple_in_the_tray` bag.

- `utils/robot_models/fetch.py:24, 53` pin empirical inside-pts validation to `apple_in_the_tray frame 488`.

Comments-in-code that name a specific frame index of a specific bag are not documentation; they are evidence that the algorithm was reverse-engineered from one episode and was never re-validated on another. A new trajectory that does not tickle these specific frame indices is not exercised by any gate; behaviour on unobserved scene structure is undefined.

The vocabulary leaks reinforce this. The detection-pipeline tests and demos all pass a fixed list shaped like `["apple", "cup", "bottle", "cabinet", "bowl"]` (`tests/demo_apis.py:218`, `tests/diagnostics/render_api_videos.py:441-442`, `tests/integration/test_detect_step_pipeline.py:10`
The per-label dynamics table covers six labels (apple, milkbox, cola, cup, pot, flowerpot) at `utils/object_dynamics.py:69-76`; everything else is silently mapped to a 5 cm spherical rigid body with e=0.40, μ=0.50 through `utils/object_dynamics.py:104-105`, with no warning emitted when the fallback is taken.

Single-frame regression artefacts are also pervasive: `FRAME_IDX = 399` hardcoded for apple_in_the_tray (`tests/diagnostics/dump_llm_relation_io.py:34`, `tests/diagnostics/visualize_llm_relation_frame.py:`
an entire test file pinned to that frame (`tests/integration/test_llm_relation_frame399.py`), and `demo_frame=281` hardcoded as the apple_drop release boundary (`tests/demo_apis.py:208`). The sole quantitative regression in the suite is `tests/parity_apple_drop.py`, a parity check against cached detections on a single bag; it cannot fail unless cache and code disagree, so it provides no signal about correctness on any other input.

IV.B Robot-platform and capture-rate hardcoding

The pipeline silently assumes a Fetch parallel-jaw gripper running at 30 Hz. URDF finger-link offsets are hard-encoded into `utils/robot_models/fetch.py:11-14` from the public Fetch hardware spec and then validated against a single bag-and-frame (`utils/robot_models/fetch.py:24, 53`). Width hysteresis values in `utils/gripper_state.py` (closed-width 25 mm, open-width 40 mm, drop-below-baseline 5 mm) are tuned for fruit-sized objects in that gripper. Time constants – birth-pending TTL of 30 frames at `ekf_tracker/config.py:99`, FSM min-transition 5 frames in `utils/gripper_state.py`, relation period 90 frames in `ekf_tracker/relations/relation_utils.py` – are interpretable only at 30 Hz. None of these constants is exposed per-platform; deploying on another robot or rate requires editing source.

IV.C Algorithmic narrowness

- **Two predict regimes are too few.** World-static and rigid-attached miss sliding, rolling, in-flight, and any deformable regime. The gravity-aware branch is a one-shot replacement at the release transition (`ekf_tracker/api.py:443`), not a process model. An object that rolls off a tray edge mid-trajectory falls back to static prediction with no further inflation.
- **Translation-only Joseph + birth-time ICP** permanently forecloses post-birth rotation refinement (`ekf_tracker/state/gaussian_state.py:546-670`). Symmetric objects (cup, bowl) are unobservable in yaw forever; an object re-encountered at a novel pose accumulates uncertainty unboundedly with no mechanism for re-alignment.
- **The keyframe freeze is correct only under stationary T_{wb}.** Loop-closure SLAM corrections retroactively shift past base poses; the frozen Σ_{bo,k_f} is then no longer the marginal posterior. The diagnostic chain at `ekf_tracker/state/obs_chain.py` is the natural substrate for retroactive correction but is never read by the fast tier.

- **Hard vs. soft label gate is two algorithms shipped under one name.** The default at `ekf_tracker/config.py:59` is `enforce_label_match=True` (hard gate); the production driver at `scripts/visualize_ekf_tracking.py:2045-2068` overrides to soft mode with β=6.0, γ=2.0, and `enforce_label_match=False`. The paper-baseline gate breaks the moment the open-vocabulary detector flips "apple" $\leftrightarrow$ "tomato" on the same physical object across frames; the production gate is what is empirically validated. The two names should not be the same algorithm.
- **Subpart suppression at** 0.8 **voxel containment** (`perception/det_dedup.py:127`) absorbs an item-shaped mask into a container-shaped mask whenever the item sits inside the container's silhouette — exactly the configuration the system claims to track via in/on relations.
- **Self-merge gate fixed at 5 cm Euclidean** (`ekf_tracker/config.py:77`) has no scale-relative variant. A 1 m cabinet collapses to a single track if SAM2 yields two masks; two adjacent items 5 cm apart in clutter survive arbitrarily.
- **Held-set closure capped at 8 iterations** (`ekf_tracker/relations/relation_utils.py:30`). Containment chains longer than 8 (item $\rightarrow$ bowl $\rightarrow$ tray $\rightarrow$ cart $\rightarrow$ shelf $\rightarrow \ldots$) are silently truncated.
- **Visibility = Fibonacci sphere of $r_{\mathbf{obj}}$** (`perception/visibility.py:59-87`) is wrong for flat (tray), elongated (broom), and non-convex objects. A flat tray seen edge-on registers near-zero p_v even when fully visible from above; the geometric occlusion does not match the logical observability the model is supposed to capture.
- **Object dynamics covers six labels** at `utils/object_dynamics.py:69-76`; everything else is silently a 5 cm spherical rigid body with e=0.40, μ=0.50. Glass, foam, soft, articulated, and large objects all inherit this default with no warning.
- **Workspace AABB fixed at** $5\times5\times3$ **m at 5 cm voxel size** (`perception/voxel_observability.py`). Tabletop scenes waste 90% of the grid; outdoor scenes truncate. Non-axis-aligned surfaces produce systematic biases in the raycast-down classification.
- **Process-noise schedule and predict-static skip-cov is hidden cross-file coupling.** The non-monotonic Q table at `utils/ekf_se3.py:347-402` (idle 1e$-$6 $\rightarrow$ unstable 1e$-$5 $\rightarrow$ stable 1e$-$8) is only safe because `ekf_tracker/state/gaussian_state.py:267` skips covariance updates for the unobserved set. Either piece in isolation is wrong; refactoring either side breaks long-occlusion behaviour with no test signal.

IV.D Bayesian incoherences

- `ekf_tracker/config.py:104` sets $r_{\text{held floor}} = 0.5$ while `ekf_tracker/config.py:52` sets $r_{\min} = 10^{-3}$. A held track survives arbitrary missed observations regardless of evidence; this is not a posterior under the miss-likelihood model ((14)), it is a hold-out. The reported existence probability for the held seed therefore loses its frequentist meaning.
- The translation-only update zeros the rotation cross-covariance rather than marginalising it (`ekf_tracker/state/gaussian_state.py:546-670`). Translation observations carry rotation information through the lever arm (Ad(T) has off-diagonal $[t]_\times R$ blocks); that information is discarded by construction. The honest rule is to set $R_{rot} = \infty$ and run the full Joseph form on the joint covariance, which would yield the same translation block but a properly marginalised rotation block. The current shortcut is a documentation hazard.

- Subpart suppression merges label histograms but discards the absorbed track's existence posterior (`perception/det_dedup.py:107-170`). A high-r small object contained inside a low-r container loses its existence weight on absorption.
- The EKF likelihood used for r_{assoc} at (13) treats the χ^2 rejection as a maximum-likelihood decision and ignores the gating prior on assignment. The Hungarian decision is therefore not coherent with the existence update under any joint posterior on (σ, r).

IV.E Evaluation vacuum

The only quantitative test in the suite is `tests/parity_apple_drop.py` — a parity match against cached detections on a single bag. There is no held-out scene, no pose-error ground-truth metric, no frame-rate stress test, no clutter scaling, no multi-robot test, no comparison against any baseline tracker. The phrase "verified to machine precision on apple_drop oid=1, fso=224" that appears in earlier documentation is a numerical sanity check on the algebraic identity (4), not a validation of the algorithm against external truth.

This documentation itself shows code/doc drift: prior to this rewrite, the LaTeX source referenced a `pose_update/` directory that has been refactored out of the repository entirely. A system whose written description does not match its source tree is, by construction, not audited end-to-end.

IV.F Top-3 worst single-scenario assumptions

1. $r_{\text{held floor}} = 0.5$ (`ekf_tracker/config.py:104`). Held tracks are pinned alive regardless of evidence. This silently keeps phantom held-object identities alive across long occlusions and turns the existence probability into a meaningless number for the held seed.
2. `LABEL_DYNAMICS_TABLE` covers six fruit-and-container labels (`utils/object_dynamics.py:69-76`). Every other label is silently mapped to a 5 cm spherical rigid body with e=0.40, μ=0.50. The gravity-aware release predict therefore degrades to a fixed-shape, fixed-restitution prior outside the validated label set with no caller signal.
3. Hidden coupling between the Q-schedule (`utils/ekf_se3.py:347-402`) and the predict-static unobserved branch (`ekf_tracker/state/gaussian_state.py:267`). The non-monotonic Q table is only safe because the unobserved branch skips the covariance update; refactoring either side in isolation breaks long-occlusion tracking with no test signal — and the only test in the suite is parity on a cached bag, so the regression cannot surface.

Part V — Per-Component Justification Audit

For each algorithmic component, exactly one of the following is given: (i) a published **Reference** that directly supports the design choice, (ii) an undeniable **Principled** reason (algebraic, information-theoretic, or Bayesian), or (iii) a **REDUNDANT** flag indicating the component is unjustified relative to its complexity, with a recommended replacement.

V.1 State and storage

Per-object Bernoulli–Gaussian belief. Reference. Vo & Vo, "The Cardinality Balanced Multi-Target Multi-Bernoulli Filter and Its Implementations" (IEEE TSP, 2009); Mahler, *Advances in Statistical Multisource-Multitarget Information Fusion* (Artech House, 2014), Ch. 7–8.

Body-frame storage with SLAM marginal decoupled. Principled. Storing μ_{bo}, Σ_{bo} in the base frame leaves the recursion independent of T_{wb}; the world view $T_{wo} = T_{wb}\,\mu_{bo}$ is the deterministic composition. SLAM updates no longer touch the per-object recursion; per-object updates no longer touch the SLAM marginal. Standard lever-arm decoupling, see Barfoot, *State Estimation for Robotics* (2017), §6–§8.

Right-trivialised SE(3) tangent (translation-then-rotation). Reference. Solà, Deray, Atchuthan, "A micro Lie theory for state estimation in robotics" (arXiv:1812.01537, 2018). The choice of right-trivialisation and ordering convention is standard for SE(3) Kalman filtering.

Dual-backend abstraction (Gaussian + RBPF). REDUNDANT. The repository ships both backends, but the RBPF tier is acknowledged buggy and the canonical instrumented driver (`scripts/visualize_ekf_tracking.py:329`) exercises only the Gaussian backend. The shared interface in `utils/base_pose_backend.py` therefore duplicates every primitive (`update_observation`, `update_observation_centroid`, `predict_static`, `predict_static_all`) into `rbpf_state.py` that no production path consumes. **Fix:** delete `ekf_tracker/state/rbpf_state.py` and `utils/base_pose_backend.py` collapse the API surface to the Gaussian backend.

`predict_static_all` alongside `predict_static`. REDUNDANT. `predict_static_all` is a one-line wrapper that forwards its arguments unchanged to `predict_static` (`ekf_tracker/state/gaussian_state` It exists only to satisfy the dual-backend interface above. **Fix:** delete with the dual-backend abstraction.

V.2 Predict

World-static motion ($\mu_{bo,k} = u_k \mu_{bo,k-1}$). Principled. Direct algebraic consequence of $T_{wb,k}\mu_{bo,k} = T_{wb,k-1}\mu_{bo,k-1}$. Identity arithmetic; no external reference needed.

Rigid-attached motion ($\mu_{bo,k} = \delta_{bg,k}\mu_{bo,k-1}$). Principled. Direct algebraic consequence of the rigid-attachment assumption (object pose constant in gripper frame).

Adjoint covariance transport. Reference. Bourmaud, Mégret, Arnaudon, Giremus, "Continuous-Discrete Extended Kalman Filter on Matrix Lie Groups Using Concentrated Gaussian Distributions" (J. Math. Imag. Vis., 2014); Forster, Carlone, Dellaert, Scaramuzza, "On-Manifold Preintegration for Real-Time Visual-Inertial Odometry" (IEEE T-RO, 2017).

Keyframe-freeze branch (skip cov update for unobserved tracks). REDUNDANT. The cross-coupling pumping that this freeze patches ($[t(u_k)]_\times P_{rr} [t(u_k)]_\times^\top$ in the Ad-conjugate) is itself an artefact of an over-large process noise on rotation. The freeze is non-Bayesian: under it, the recursion's posterior at frame k is no longer the marginal of any joint posterior, and any retroactive SLAM correction (loop closure) is silently inconsistent with the frozen Σ_{bo,k_f}. **Fix:** set $Q_{\text{static}} = 0$ for genuinely static objects (no random walk, no scheduled inflation) and run the recursion uniformly. The cross-coupling is then driven only by genuine motion.

Q schedule on `frames_since_obs` (1e−6 → 1e−5 → 1e−8). REDUNDANT. Adaptive Q on observation age has no Bayesian justification. The non-monotonic ramp (small idle / larger "unstable" / smaller "stable") is tuned to mimic the keyframe-freeze behaviour approximately; it is a workaround for a workaround. Maybeck, *Stochastic Models, Estimation, and Control* (1979), §1.5: Q should reflect physical motion uncertainty, not estimator state. **Fix:** one Q per regime (static / manipulation), no schedule.

Covariance saturation Φ (trace cap on P). **REDUNDANT.** The default `P_max=None` (`ekf_tracker/config.py:55`) means the cap is dead code in production. Trace-clipping violates posterior optimality and consistency (Maybeck §1.5). **Fix:** delete `saturate_covariance` and the `P_max` kwarg. If covariance grows unboundedly, repair the predict / observation models, do not clip.

V.3 Detection conditioning

Voxel-containment subpart suppression. **Principled.** 3D analogue of 2D non-maximum suppression. A small mask whose voxel set is mostly inside a larger mask's voxel set is, under rigid-body assumptions, almost certainly an over-segmentation of the larger mask. Compare 2D NMS in Neubeck & Van Gool, "Efficient Non-Maximum Suppression" (ICPR, 2006).

Centroid validity gate. **Principled.** A mask without valid-depth pixels carries zero 3D information; dropping it is the only honest behaviour.

V.4 Association

Hungarian assignment. **Reference.** Munkres, "Algorithms for the Assignment and Transportation Problems" (J. SIAM, 1957); Bar-Shalom & Fortmann, *Tracking and Data Association* (Academic Press, 1988), §5.

χ^2_3 probabilistic gate at $G_{\mathbf{out},t} = 21.108$. **Reference.** Bar-Shalom, Li, Kirubarajan, *Estimation with Applications to Tracking and Navigation* (Wiley, 2001), §3.3. χ^2_n tail at 99.97% is a standard MIPDA outer-gate.

Translation-only Mahalanobis (assume $R_{rot} = \infty$). **Principled.** Equivalent to marginalising rotation out of the joint χ^2_6. The translation-block residual under linear-Gaussian assumptions is exactly χ^2_3.

Hard Euclidean residual cap (`max_residual_m`=0.30). **REDUNDANT.** This second gate is admitted in the source comment to catch the "inflated-covariance pathology" — it patches a failure of the χ^2 gate caused by the keyframe-freeze and Q-schedule machinery (V.2). Two gates with one principle is one too many. **Fix:** repair Σ_{bo} first (V.2); then remove the cap and let the χ^2 gate be the sole probabilistic admission criterion.

SAM2 continuity bonus $\alpha = 4.4$. **REDUNDANT.** The mechanism (cost discount for matched perception ids) is plausible but the value 4.4 has no principled derivation. The post-hoc "χ^2_3 at 78%" interpretation is unmotivated. **Fix:** replace with a likelihood factor $\log p(\tau_o = \text{pid}_d)$ derived from SAM2's empirical id-stability statistics, or remove and rely on the χ^2 gate alone. Encoding identity continuity as a likelihood places it in the same probabilistic framework as the χ^2 gate.

Hard label gate (default `enforce_label_match=True`). **REDUNDANT.** The hard gate is the $\beta = \infty$ limit of the soft penalty (next item). Two cost-function formulations for one purpose; the production driver overrides the default to soft mode, so the hard mode is a non-default that no shipped code path actually exercises in regression. **Fix:** keep the soft formulation only. Set β large (but finite) when behaviour-equivalent to "hard" is wanted.

Soft label penalty β and score weight γ. **Principled.** Augmenting the assignment cost with a label-mismatch penalty and a low-score penalty is a likelihood-style construction. The specific values ($\beta = 6.0$, $\gamma = 2.0$ in the production driver) are unjustified constants but the mechanism is sound, provided one formulation (this one) is canonical.

V.5 Update

Joseph form on full SE(3). **Reference.** Bar-Shalom, Li, Kirubarajan (2001), §5.3; Maybeck (1979), §6.2. PSD by construction.

Translation-only Joseph with $P_{rt}^{\mathbf{post}} = 0$. **REDUNDANT.** The shortcut zeros the rotation cross-covariance instead of marginalising it. Setting $R_{rot} = \infty$ in the full Joseph update yields the same translation block *and* a properly marginalised rotation block via the Schur complement (Maybeck, §6.2; Crassidis & Junkins, *Optimal Estimation of Dynamic Systems*, 2011, §3.5). The current path discards the lever-arm-coupled rotation information that translation observations actually carry through the off-diagonal $[t]_\times R$ blocks of $\mathrm{Ad}(T)$. **Fix:** use the full Joseph form with R_{rot} a large finite number. The rotation block then updates correctly and the code path is unified with the ICP update.

Held-track prefilters (`held_meas_radius_m`=0.25, `held_meas_innov_max_m`=0.20). **REDUNDANT.** Two ad-hoc Euclidean gates added on top of the χ^2 gate, the residual cap, and the held-mode Q. Four overlapping mechanisms guard one update. The principled mechanism is the held-mode Q schedule (large during grasping/releasing, small during holding) which already encodes the physically expected innovation. **Fix:** remove the prefilters. Trust the Q schedule and the χ^2 gate.

One-shot ICP at admission. **Principled.** Pragmatic compute budget: ICP per frame is expensive, ICP once at birth gives a reasonable rotation prior. The known weakness (no rotation refinement post-admission) is documented and acknowledged. Reference: Besl & McKay, "A Method for Registration of 3-D Shapes" (IEEE PAMI, 1992).

Three ICP backends (`centroid` / `icp_chain` / `icp_anchor`). **REDUNDANT.** Default is `icp_chain`; the other two are documented as "what the original integration test used" and "strict decomposition" respectively, and neither is exercised by any production test. **Fix:** keep `icp_chain`; move `centroid` and `icp_anchor` to a dedicated `tests/icp_backends/` fixture, or delete.

Huber inner / outer reweight. **Reference.** Huber, "Robust Estimation of a Location Parameter" (Annals of Math. Stat., 1964); Zhang, "Parameter Estimation Techniques: A Tutorial" (Image & Vision Computing, 1997).

Adaptive Barron robust loss (slow tier only). Reference. Chebrolu, Magistri, Läbe, Stachniss, "Adaptive Robust Kernels for Non-Linear Least Squares Problems" (IEEE RA-L, 2021). Cited explicitly in the module header `perception/adaptive_kernel.py:2`.

World-frame composition μ_{wo}, Σ_{wo} ((9)). Principled. First-order propagation of the joint (T_{wb}, μ_{bo}) uncertainty under deterministic composition, treating T_{wb} and μ_{bo} as independent. Standard lever-arm arithmetic; see Barfoot (2017), §7.5.

Dual update primitives (`update_observation` vs `update_observation_centroid`). REDUNDANT. Two implementations of the same primitive: full-SE(3) Joseph and translation-only Joseph. The centroid form is the special case of the full form with $R_{rot} \to \infty$. The duplication is also mirrored into `rbpf_state.py`, giving four implementations of one primitive. **Fix:** implement once in the full form; the centroid case is recovered by passing R_{rot} large.

V.6 Birth and prune

Bernoulli existence updates (r-predict, r-assoc, r-miss). Reference. Mahler (2014), Ch. 7–8; Vo, Vo, Hoseinnezhad, "Robust Multi-Bernoulli Filtering" (IEEE J. STSP, 2013).

Birth existence $r_0 = \lambda_b \hat{s}/(\lambda_b \hat{s} + \lambda_c)$. Reference. Vo & Vo (IEEE TSP, 2009). Standard CB-MeMBer birth.

Confirm-k M-of-N gate. Reference. Bar-Shalom & Fortmann (1988), §3. M-out-of-N detection is classical track-confirmation.

Confirm-k=3, score-floor=0.20, border-margin=2 px, TTL=30 frames, near-live=0.05 m. REDUNDANT. Magic numbers tuned for one trajectory's image resolution, capture rate, and object scale. Border 2 px assumes a 640×480 stream; TTL=30 frames assumes 30 Hz capture; near-live=5 cm assumes fruit-sized objects. **Fix:** replace with relative gates — border = 1% of image width, TTL = N seconds at any rate, near-live = $k\sigma_t$ on the joint translation cov.

$r_{\min} = 10^{-3}$ prune threshold. Principled. Standard Bernoulli filter pruning on a memory / precision tradeoff. The order of magnitude follows directly from the floating-point sensitivity of (13).

$r_{\textbf{held floor}} = 0.5$. REDUNDANT. Mathematically incoherent. The held seed's posterior r should be governed by (14); if the model says r should fall below 0.5 given the miss-likelihood evidence, it should fall. Pinning r above 0.5 converts it into a non-probability. **Fix:** remove. If a held seed must survive long occlusions, encode the prior via $p_d \to 0$ during full occlusion (which is exactly what the visibility module is for) rather than by floor-pinning the posterior.

V.7 Self-merge

Information-sum fusion of two Gaussians. Reference. Mutambara, *Decentralized Estimation and Control for Multisensor Systems* (CRC Press, 1998), §4.3. Approximate (independence assumed) but the standard track-to-track fusion form.

Same-label gate at fixed 5 cm Euclidean (`self_merge_trans_m`=0.05). REDUNDANT. Fixed-distance, no scale-relative variant. Two adjacent fruit at 5 cm survive arbitrarily; two SAM-split masks of one cabinet collapse arbitrarily. **Fix:** replace with $d_t^2 \le \chi_3^2(0.95) = 7.815$ on the joint translation cov $P_{tt,o} + P_{tt,d}$.

Protected-pairs gate (relations-derived $\mathcal{P}$). Principled. If the scene graph asserts two oids are physically distinct (e.g. apple on tray), they must not collapse. Direct logical consequence of the relation semantics.

V.8 Visibility

Continuous p_v in miss-frame likelihood ($\tilde{p}_d = p_d \cdot p_v$). Principled. Standard Bernoulli filter convention under known visibility (Mahler, 2014, §7.6).

Fibonacci-sphere visibility predicate. REDUNDANT. The principled visibility test is a depth-test against the per-object TSDF / SDF (Newcombe et al., "KinectFusion: Real-Time Dense Surface Mapping and Tracking," ISMAR 2011). The spherical fallback is wrong for any non-spherical object: a flat tray seen edge-on registers near-zero p_v even when fully visible from above. **Fix:** implement TSDF-projection visibility per-object, or demote the visibility module to a binary in-frustum test only.

V.9 Relations

Per-edge EMA smoothing ($\alpha = 0.3$, threshold 0.5). Principled. First-order low-pass to reject single-frame backend flicker. The mechanism (any digital-filter text) is sound; the specific α is a time-constant choice but does not affect correctness.

Trigger gate (event triggers + every-90-frame heartbeat). REDUNDANT. The event-trigger set (first call, release transitions, new-oid arrivals) covers every state change that could alter the scene graph. The 90-frame periodic backstop is a hedge against unspecified event-coverage gaps. **Fix:** drop the heartbeat. If the event set under-covers, audit the event set instead of patching with periodic re-fires.

Convention swap (parent=supported, child=supporter, against the LLM's j-on-i). REDUNDANT. Exists to fix a sign-error regression. Two conventions mixed in one codebase, with a swap at every emission point. **Fix:** canonicalise once at the LLM client boundary — the output convention should match held-set expansion's parent/child semantics directly. No swap.

V.10 Held-set expansion

Closure under in/on edges. Principled. Held-set semantics: anything supported by the held seed is also held. Direct logical consequence.

Iteration cap of 8. REDUNDANT. A fixed cap on a fixpoint iteration cannot accommodate arbitrary chain depth. Containment chains in real scenes can exceed 8 (item-on-bowl-on-tray-on-cart-on-shelf-...). **Fix:** run the closure to convergence with a `while changed:` loop. Worst case is $O(N \cdot |\mathcal{E}|)$, still trivial.

V.11 Gravity-aware release

Voxel observability grid (3-state with hit / pass hysteresis). Reference. Hornung, Wurm, Bennewitz, Stachniss, Burgard, "OctoMap: An Efficient Probabilistic 3D Mapping Framework Based on Octrees" (Auton. Robots, 2013). The 3-state model is a coarse simplification of OctoMap's log-odds occupancy update with explicit hysteresis instead of log-odds clamping.

Voxel-driven h branches `hit_occupied` and `all_empty`. Principled. Both have unambiguous geometric semantics: a clean surface terminates the column at a known $h^\star$; a fully-empty column terminates at the workspace floor.

Voxel-driven h branches `mixed_unseen` extra inflation and `all_unseen` midpoint heuristic. REDUNDANT. The midpoint choice for `all_unseen` is an arbitrary compromise between worst-case and best-case; the `mixed_unseen` "path-uncertainty" inflation $(h_u - h^\star)/2$ has no derivation. **Fix:** for `all_unseen`, return a flat uninformative prior over the reachable depth range (uniform distribution lifted to the EKF as a Gaussian with matching first two moments). Drop the `mixed_unseen` extra inflation.

Object dynamics table $(e, \mu, \mathrm{shape}, r_{\mathbf{obj}})$ for six labels (`apple`, `milkbox`, `cola`, `cup`, `pot`, `flowerpot`). REDUNDANT. Six labels, values cited as "from agricultural / engineering literature" with no per-label paper, plus a silent fallback for every other label. Brittle to ontology drift; the dynamics-conditioned σ field of the gravity prior contributes at most one frame of covariance and is then either confirmed or overwritten by subsequent observations. **Fix:** replace the table with one conservative prior $P_{\mathrm{drop}} = \mathrm{diag}((10\,\mathrm{cm})^2{\times}3,\ (\pi/2)^2{\times}3)$ (or similar). Let the EKF re-converge from observations.

Parametric dispersion model ((17): $\sigma_{xy}, \sigma_z, \sigma_{\mathbf{yaw}}$ in terms of h, e, $r_{\mathbf{obj}}$, $v_{\mathbf{rel}}$). REDUNDANT. The functional forms are invented: the bounce term $\sigma_{\mathrm{bounce}} \propto v_{\mathrm{impact}}/(1-e)$ has no impact-mechanics citation; the yaw term $\sigma_{\mathrm{yaw}} = \pi(1 - \exp(-h(1-e^2)/r_{\mathrm{obj}}))$ is fitted, not derived. The constant $\varepsilon_{\mathrm{rough}} = 5 \cdot 10^{-3}$ carries the source comment "tune empirically on the apple_drop trajectory" — an explicit admission that the value is fitted on one trajectory. **Fix:** delete the dispersion model. Replace with the constant prior above. The EKF posterior after two or three subsequent frames of valid detection is indistinguishable from the dispersion-model output, and this version has no scenario-tied tuning.

One-shot release-trigger (Releasing→Idle, overwrite EKF state once). REDUNDANT. A proper Bayesian filter has a FALLING/SETTLING regime that runs every frame between holding and re-observation. The one-shot is a hack to avoid implementing a continuous ballistic predict. **Fix:** add a FALLING predict to the predict step (ballistic motion under gravity until the predicted impact frame, then a transition to the constant-prior state); remove the one-shot overwrite.

Workspace AABB fixed at $5 \times 5 \times 3$ m, 5 cm voxel. REDUNDANT. Fixed-size hardcoded grid wastes 90% of the volume on tabletop scenes and truncates outdoor scenes. **Fix:** derive the AABB from the SLAM map's bounding box plus a margin; pick voxel size from r_{obj} (e.g. $r_{\mathrm{obj}}/4$) at allocation time.

V.12 Gripper FSM

Four-phase FSM with width hysteresis and stability test. Principled. Standard FSM with debouncing (Åström & Murray, *Feedback Systems*, Princeton 2008, §9). Hysteresis is the textbook anti-flicker mechanism for a threshold-driven discrete state.

Width thresholds (closed=25 mm, open=40 mm, delta=5 mm, min-frames=5). REDUNDANT. Tied to one robot's gripper geometry and one frame rate. **Fix:** derive thresholds from the URDF maximum width as percentages (closed = 25% of max, open = 50%); convert `min_transition_frames` to a time constant (150 ms) and divide by the frame rate.

Held-oid selection by inside-gripper containment fraction. Principled. The mask whose 3D points are most-contained in the gripper finger volume is, by direct geometric inference, the most likely held object. The smallest-mask-area tiebreak is also principled (small held items are harder to over-segment than large background items).

V.13 Codebase-level redundancies

`perception/camera_pose_refiner.py` (337 LoC). REDUNDANT. Imported only by `scripts/data_demo.py` and `scripts/realtime_app.py`, both legacy. Imports `heuristic_tracker.scene_object` from the heuristic tracker that the EKF replaced. **Fix:** delete the module and its two callers.

Slow tier in `ekf_tracker/factor_graph.py` (when not promoted). Principled. The slow tier is a factor-graph smoother that fires only when explicitly promoted. As an out-of-band module this is acceptable; it is not on the production fast-tier critical path.

`ekf_tracker/manipulation/gripper_state_inferrer.py` (45-line shim). Principled. Driver-side wrapper that adapts `utils/gripper_state.py:GripperPhaseTracker` to the tracker-coupling shape used by the EKF. Acceptable separation of concerns; the FSM logic itself is not duplicated.

V.14 Tally and recommendation

Of the 58 components catalogued above:

- **13** are **Reference**-justified (Bernoulli MTT literature, Bar-Shalom and Mahler, Joseph form, Munkres, Huber, Barron / Chebrolu, OctoMap).
- **19** are **Principled** (algebraic, geometric, or Bayesian consequences with no useful citation beyond the obvious).
- **26** are **REDUNDANT**.

  The redundancies cluster around four loci:

1. **Predict patches (V.2):** the keyframe-freeze / Q-schedule / saturation triple together patch a problem that would not exist under an honest fixed Q.
2. **Update overlap (V.4–V.5):** four overlapping admission gates (`max_residual_m`, `held_meas_radius_m`, `held_meas_innov_max_m`, plus the χ^2 gate) and a trans-only Joseph that discards rotation information instead of marginalising it.
3. **Gravity-predict subsystem (V.11):** an entire 200+ LoC subsystem ($\sigma_{xy}, \sigma_z, \sigma_{\text{yaw}}$, label-conditioned restitution table, four-state column classification) that contributes one frame of covariance and is empirically tuned on `apple_drop`.

4. **Magic-number ribbon (V.6 / V.12):** fixed birth, FSM, workspace, and self-merge constants that tie the system to one trajectory's image resolution, capture rate, robot platform, and object scale.

Removing the 26 redundant items is estimated to simplify the codebase by 30–40% (the gravity-predict subsystem alone is a quarter of the redundancy mass) without changing any production-validated behaviour, since none of the redundant items is exercised by an external regression test that is not equally satisfied by its proposed replacement.